Theoretical Basis from String Theory Which Precludes Brain Quantum Computation

Abstract

We present a theoretical framework grounded in string theory that precludes the possibility of brain-based quantum computation via gravitational modulation. Higher-order corrections to the worldsheet beta function, typically considered negligible in conventional physics, lead naturally to $f(R)$ gravity in the low-energy effective action. These corrections alter the gravitational wave spectrum by introducing high-frequency scalar curvature modes. As neural systems exhibit low-pass filtering, these high-frequency components become biologically imperceptible. Consequently, action potential timing in axons remains unmodulated, ruling out gravitationally-induced quantum computation in the brain.

1 Introduction

Our 2021 study demonstrated that gravitational waves in standard General Relativity (GR) can modulate action potential timing in axon tracts by altering axonal projection lengths through spacetime oscillations. In string theory, however, gravitational dynamics are determined by beta functions arising from conformal invariance on the worldsheet. These beta functions contain higher-order curvature corrections, often discarded as small, but fundamentally quantum in nature.

Here, we show that including these corrections transforms the gravitational theory into $f(R)$ gravity, suppressing biological sensitivity to gravitational waves by upshifting frequency content beyond neural response ranges. As a result, neural systems behave as if gravitational waves do not exist.

2 f(R) Gravity and Frequency Filtering

Higher-order beta function corrections—such as R^2, R^3, and beyond—lead naturally to modified gravity theories like $f(R)$. These generate scalar curvature wave modes that obey a Klein-Gordon equation, leading to gravitational wave spectra with upshifted frequency components.

Axon tracts and other neural structures act as low-pass filters, unable to transduce these higher frequencies. Consequently, there is no oscillatory modulation of interaxonal distances, no delay variation in action potentials belonging to tracts of coupled axons, and no neural sensitivity to these modified gravitational waves.

3 Quantum Localization Duality

A deeper insight emerges: if the full quantum structure of spacetime is accounted for via beta function corrections, then gravitational modulation of the brain becomes impossible. Conversely, if one suppresses these quantum corrections, gravitational effects on the brain may appear, but the theory is no longer quantum-complete.

Quantum Exclusivity Principle of Biophysical Gravity: In any consistent string-based framework, quantum behavior can manifest either in spacetime structure (via beta corrections) or in neural systems (as quantum computation), but not both simultaneously.

This introduces a form of quantum localization, whereby the system's quantum behavior is constrained to a specific domain—geometry or biology—but cannot simultaneously exist in both.

4 Conclusion

String theory naturally leads to modified gravity with suppressed biological effects. The inclusion of higher-order corrections in beta functions leads to gravitational wave filtering and consequently inhibits brain-based quantum computation. Our findings suggest a deep theoretical duality—either quantum corrections belong to the geometry or to the brain, but not both. This introduces a new gravitational epistemology with implications for consciousness studies, quantum neuroscience, and high-energy biophysics.

Visual Abstract